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Report on Testing Methods: EN 17152-1 vs. ASTM D2412

Introduction

This report provides a comparative analysis of the testing methods specified in **EN 17152-1:2019** and **ASTM D2412**. These standards are used to evaluate the performance and durability of plastic piping systems for underground conveyance and storage of non-potable water.

EN 17152-1:2019

Purpose

EN 17152-1:2019 specifies the requirements for plastic piping systems used for the underground conveyance and storage of non-potable water, such as stormwater.

Materials

- Polypropylene (PP)
- Unplasticized polyvinyl chloride (PVC-U)

Applications

- Infiltration
- Attenuation
- Storage of non-potable water

Testing Procedures

1. **Short-Term Compression Strength**: Determines immediate load-bearing capacity.
2. **Long-Term Compression Strength**: Assesses performance under sustained loads over time.
3. **Material Tests**: Verifies properties of materials used.

Short-Term Tests

1. **Compression Strength Test**
   - **Objective**: Assess immediate load-bearing capacity.
   - **Procedure**: Apply compressive load until deformation or failure occurs.
2. **Impact Resistance Test**
   - **Objective**: Evaluate resistance to impacts during installation or service.
   - **Procedure**: Subject the box to impacts and inspect for damage.
3. **Flexural Strength Test**
   - **Objective**: Measure resistance to bending forces.
   - **Procedure**: Apply load at the center of the box until deformation or failure.

ASTM D2412

Purpose

ASTM D2412 is the Standard Test Method for Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate Loading.

Testing Procedure

1. **Parallel-Plate Loading**

- **Objective**: Determine load-deflection characteristics and pipe stiffness.
- **Procedure**: Apply compressive load at a constant rate (2.0% of chamber height per minute) until specified deflection or failure occurs.

Key Differences and Implications

Load Application and Rate

- **EN 17152-1**: Focuses on immediate load-bearing capacity, impact resistance, and flexural strength.
- **ASTM D2412**: Uses a controlled loading rate to determine load-deflection characteristics and stiffness.

Material Specifications

- **EN 17152-1**: Covers PP and PVC-U materials.
- **ASTM D2412**: References additional standards for PP and PE materials.

Testing Environment

- **EN 17152-1**: Simulates real-world scenarios.
- **ASTM D2412**: Conducted in a controlled laboratory environment.

Performance Prediction

- **EN 17152-1**: Assesses immediate stresses for installation and initial use.
- **ASTM D2412**: Evaluates long-term performance under sustained loads.

Implications

- **Installation and Handling**: EN 17152-1 ensures boxes can withstand stresses during installation.
- **Long-Term Performance**: ASTM D2412 provides insights into long-term performance.
- **Material Suitability**: ASTM D2412's referenced standards ensure materials meet specific requirements.

Conclusion

In conclusion, both **EN 17152-1** and **ASTM D2412** testing methods are designed to ensure the durability and performance of plastic piping systems, but they focus on different aspects. **EN 17152-1** emphasizes immediate load-bearing capacity, impact resistance, and flexural strength, making it suitable for assessing performance during installation and initial use.
On the other hand, **ASTM D2412** focuses on load-deflection characteristics and pipe stiffness, providing valuable insights into long-term performance under sustained loads.
The choice between these methods depends on the specific requirements of the project, and in some cases, using both methods together could offer a more comprehensive evaluation.

Regards,

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Attachments: Summary of EN-17152-1 & ASTM D2412 AquaCell testing;

COMPRESSION TESTING COMPARISON (EN v ASTM)		
TOP SIDE	EN-17152-1	ASTM D2412
Standard Configuration (SD)	64.35 psi	70.0 psi
Extra Strong Configuration (EX)	100.03 psi	106.1 psi

LONG SIDE	EN-17152-1	ASTM D2412
Standard Configuration (SD)	16.29 psi	17.2 psi
Extra Strong Configuration (EX)	28.33 psi	26.4 psi

AQUACELL COMPRESSION TEST RESULTS (ASTM D2412) - APRIL 23, 2024 & DECEMBER 19, 2024		
Top Side	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	77,779 lbf	117,839 lbf
Average Compression Strength (psi)	70 psi	106.1 psi

Long Side	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	13,645 lbf	22,453 lbf
Average Compression Strength (psi)	17.2 psi	26.4 psi

AQUACELL COMPRESSION TEST RESULTS (EN 17152-1) - COMPILED APRIL 19, 2021		
Top Side	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	72,089 lbf	111,580 lbf
Average Compression Strength (psi)	64.35 psi	100.03 psi

Top Side 2 Layers	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	59,949 lbf	NA
Average Compression Strength (psi)	53.71 psi	NA

Long Side Top	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	24,279 lbf	NA
Average Compression Strength (psi)	16.29 psi	NA

Long Side Bottom	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	15,699 lbf	NA
Average Compression Strength (psi)	20.31 psi	NA

Small Side Top	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	13,773 lbf	NA
Average Compression Strength (psi)	18.52 psi	NA

Small Side Bottom	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	6,392 lbf	NA
Average Compression Strength (psi)	16.56 psi	NA

Long Side	Standard Configuration (SD)	Extra Strong Configuration (EX)
Average Compression Strength (lbf)	NA	24,279 lbf
Average Compression Strength (psi)	NA	28.33 psi